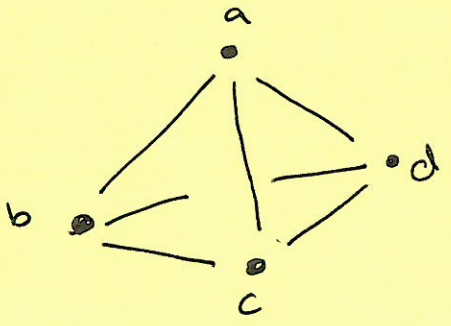
8 NOV 2023

<u>Lemma</u>: A graph $G=(V,E)$ has an Eulerian tour <u>iff</u> there are exactly 0 or 2 vertices of odd degree.

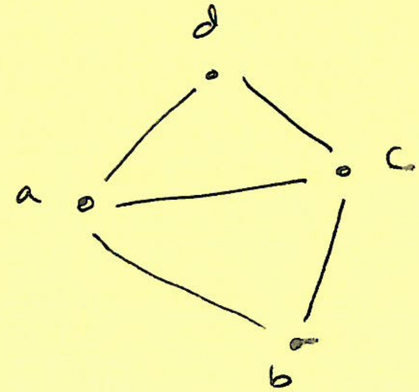
→ If 2: these are the verts to start/end at

→ If 0: can start anywhere. Note, in this case, the tour is a circuit (starts + ends at same place)

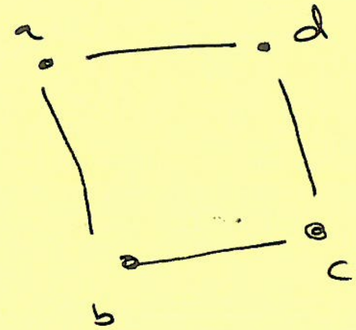
<u>Examples</u>: Is an Euler tour (a walk using every edge <u>exatly</u> once) possible:



NO!



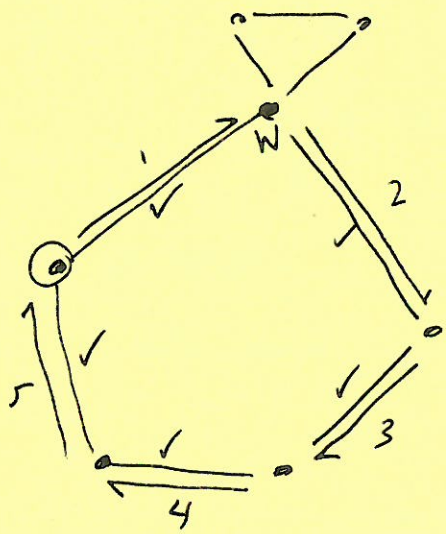
YES!
We must start at either a or c.



YES!
In fact, there are 8 such walks.

INPUT: a graph $G = (V, E)$ such that there are no vertices of odd degree.

OUTPUT: An Eulerian tour of G.



Algorithm: Hierholzer's Algorithm

1. Pick $v_0 \in V$ as start vertex
1b. $i = 1$; v_i = nbr of v. Mark edge (v_0, v_1)
2. While $v_i \neq v_0$

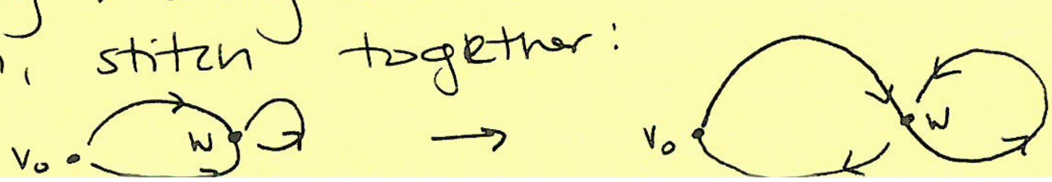
    $v_{i+1} \leftarrow$ a nbr of v_i s.t. edge is not marked yet (✓)

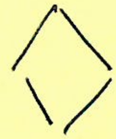
    mark (v_i, v_{i+1})

    i++

   endwhile
3. Find w on my cur path that has an unmarked edge going out.
   follow same procedure to find a path starting + eding at w
   Then, stitch together:

   } repeat until no more unmarked edges.





<u>Lemma</u>: If there is an unmarked edge, then at least one unmarked edges use vertices of marked edges.